

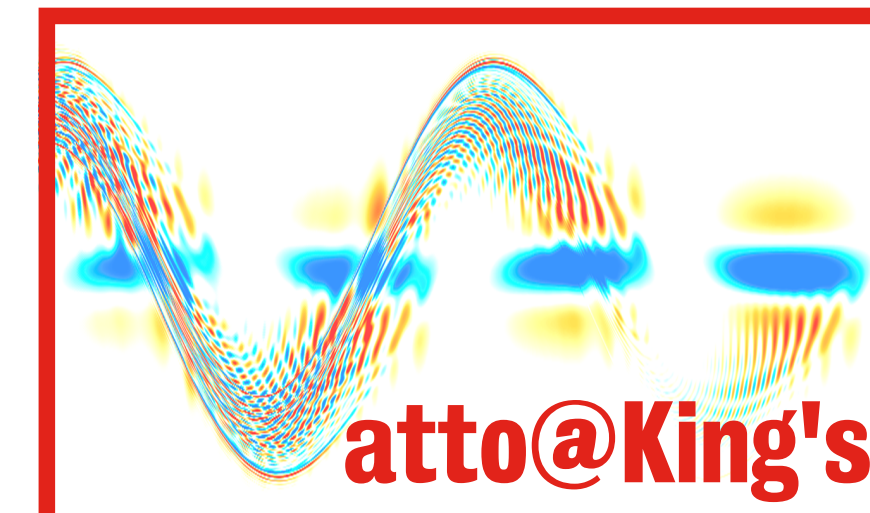
SNAIL : A library for fast SFA calculation of HHG response in gases

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Welcome to SNAIL! SFA Numerically Accelerated Integration Library (SNAIL)

- Calculates simple HHG spectra with a few lines of code
- Quickly generates large datasets for Machine Learning
- Try out new field shapes just to see what happens!



All the code in this poster is the minimum required to produce the graphs shown. To see more examples and try out the code, go to:
<https://github.com/atto-King-s/SNAIL>

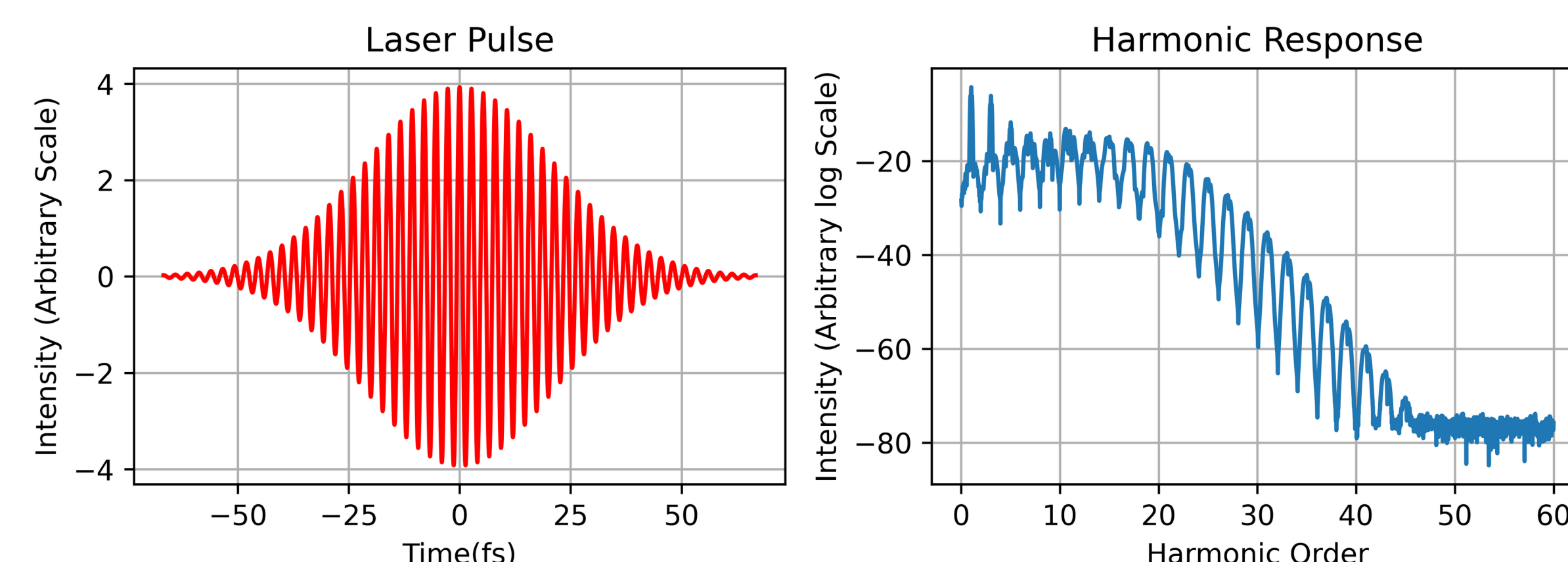
Xenon Gas Target Response

SNAIL is designed for ease of use at both small and large scale. A simple example, calculating the HHG response of a xenon atom driven by a Ti:Sapphire laser. Leveraging the use of presets, the following code outputs the full HHG spectrum.

```
import general_tools as gt
import plotting_tools as pt

config = gt.Tilaser
field = gt.generate_pulse(config)
resp = gt.dipole_response(field, config)
pt.driving_spectra(resp, field, style='h')
```

0.5 seconds



Cutoff Law Verification

To check the physicality of SNAIL, we can check the cutoff law [1]. The plateau in a harmonic spectrum ends at a harmonic given by:

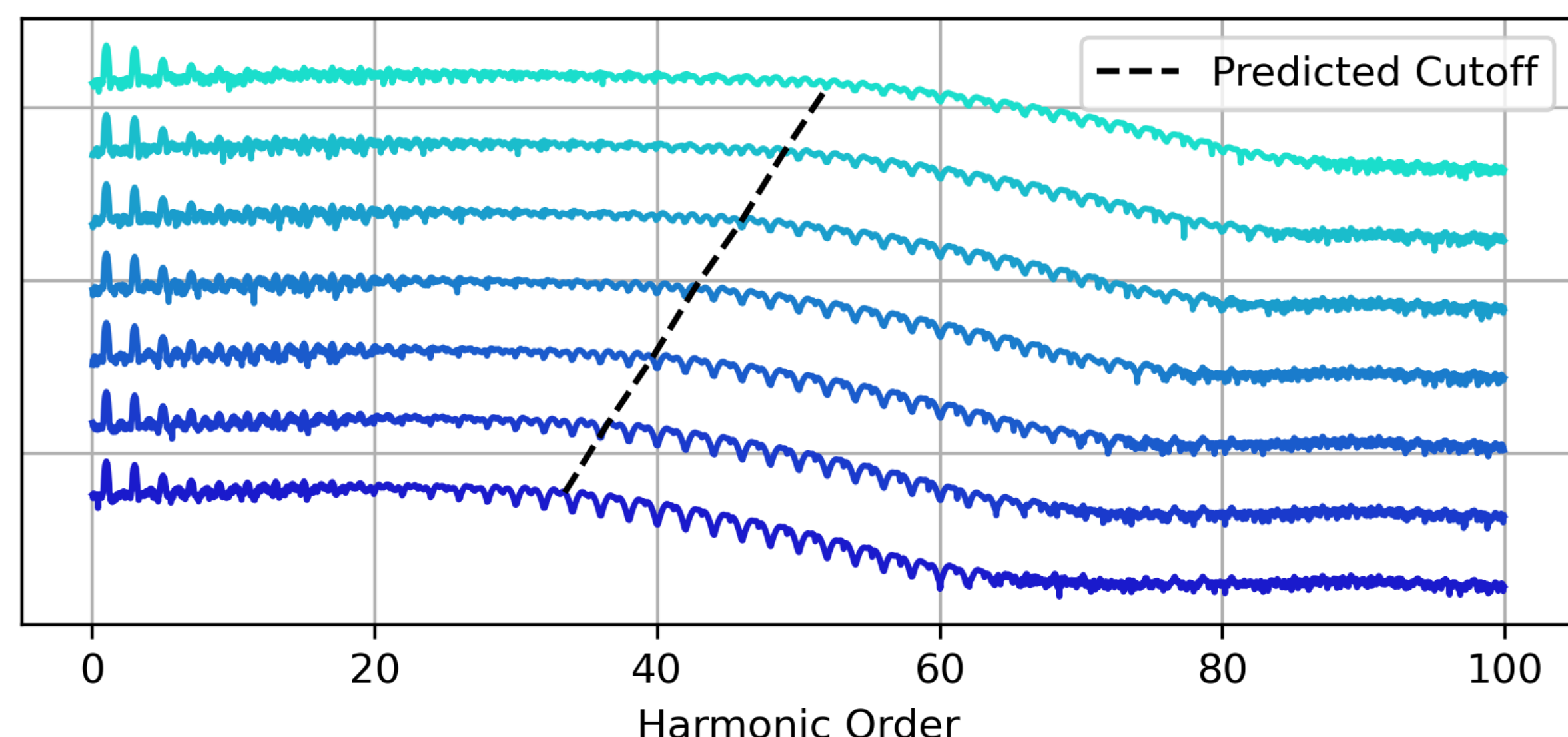
$$E_{max} = I_p + 3.17U_p$$

By varying the intensity and the ionization potential, we can check it follows a linear relation:

```
config = gt.Tilaser
cutoff = []
for i in range(7):
    config.Ip += i*0.3
    config.peak_intensity += 0.12e14*i
    field = gt.generate_field(config)
    resp = gt.dipole_response(field, config)
    cutoff.append(cutoff, gt.cutoff(config))
    pt.harmonic_plot(resp + 40*i)

final = np.array(cutoff)
plt.plot(final[:,0], final[:,1], 'r--')
```

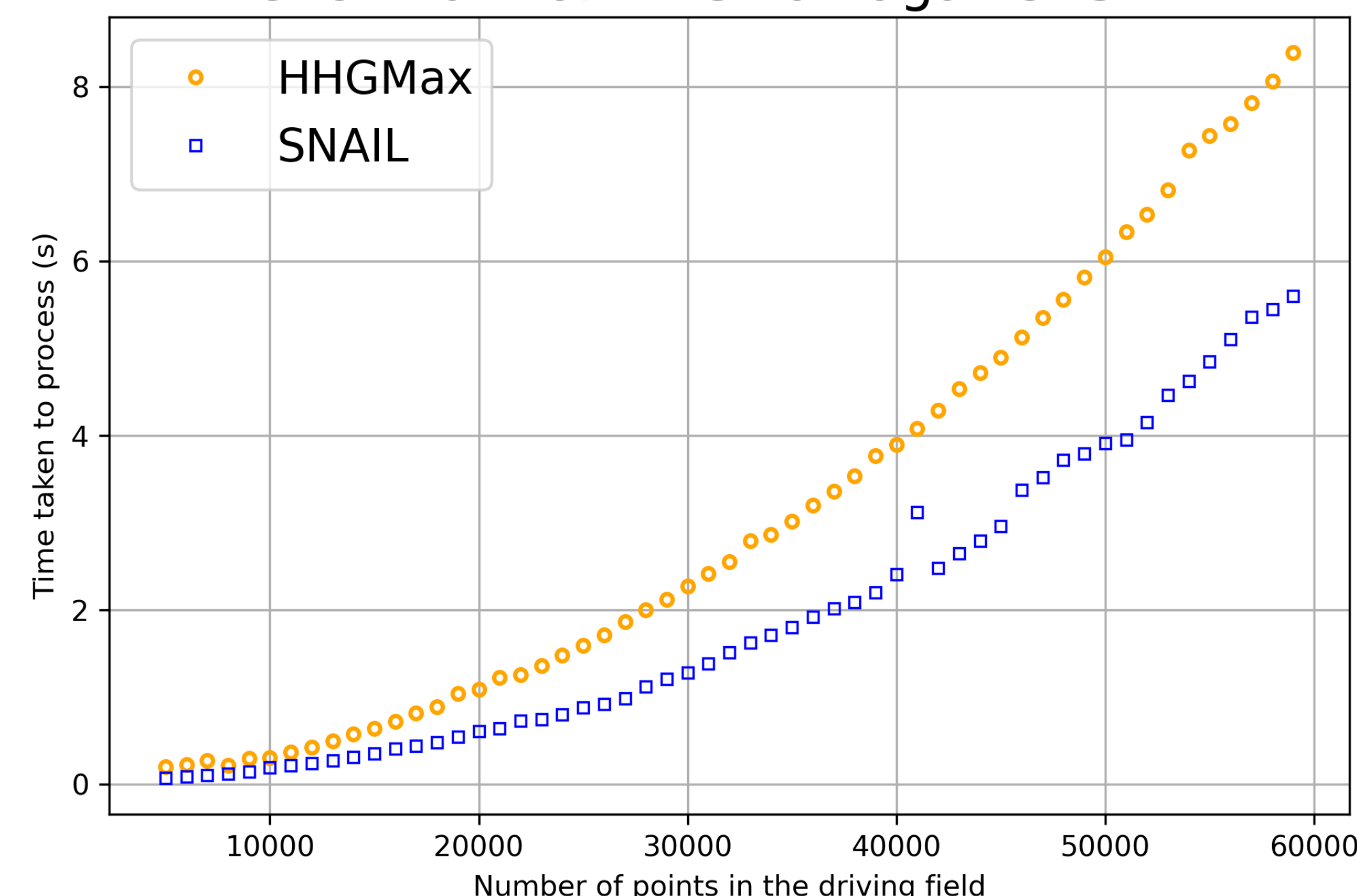
Cutoff Law Demonstration With Simulated Data



Benchmarking

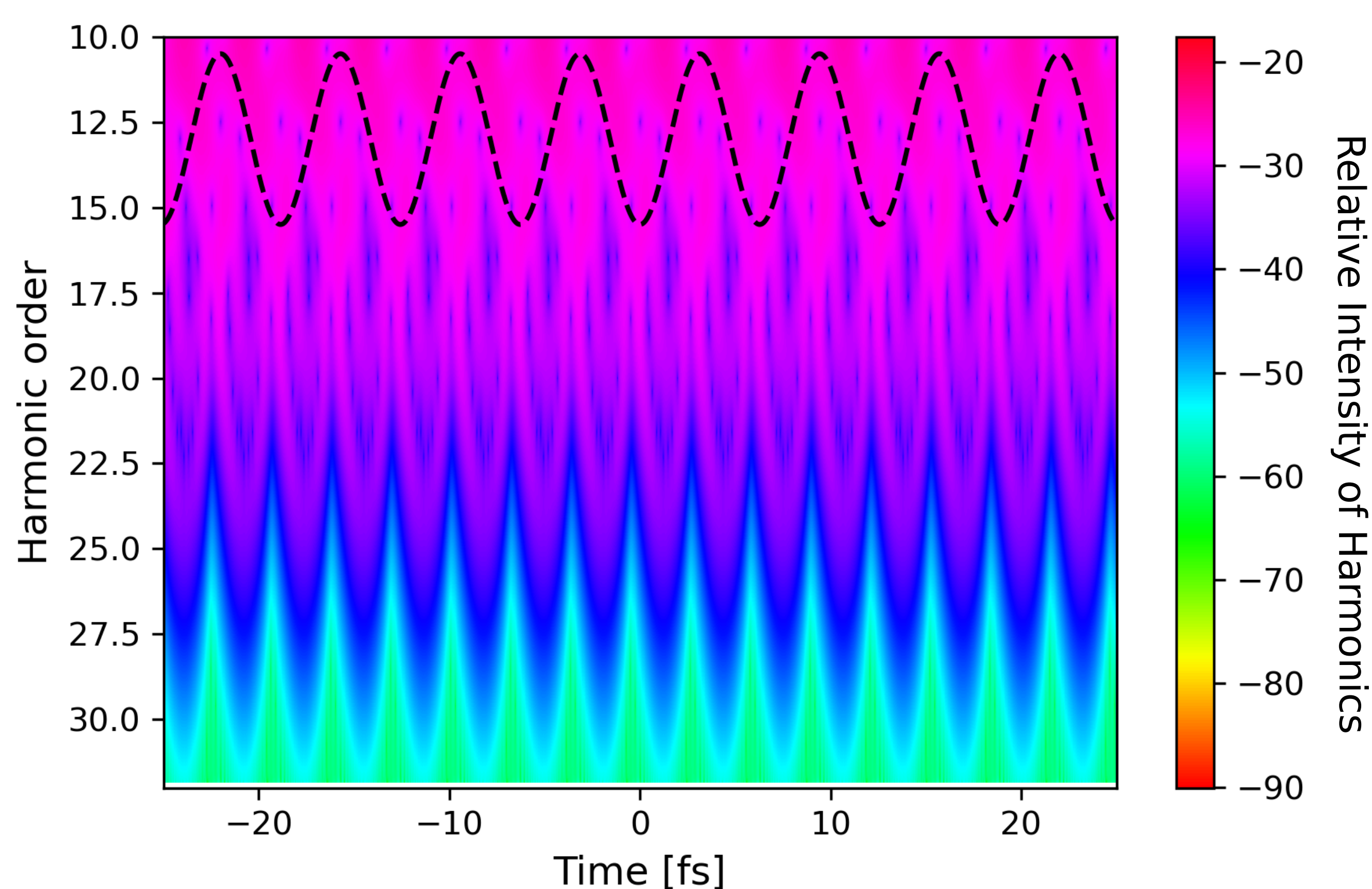
We've seen that it works, but is it fast? The best way to test this is to compare it against the fastest competitor, HHGMax [2]. The benchmark was done on 10 runs for each number of driving field points, and on a rebooted computer with nothing else running. SNAIL used the parallelisation option. Despite HHGMax being written in C++, SNAIL outpaces it at higher point counts, which quickly adds up for large datasets.

Benchmark of HHGMax against SNAIL



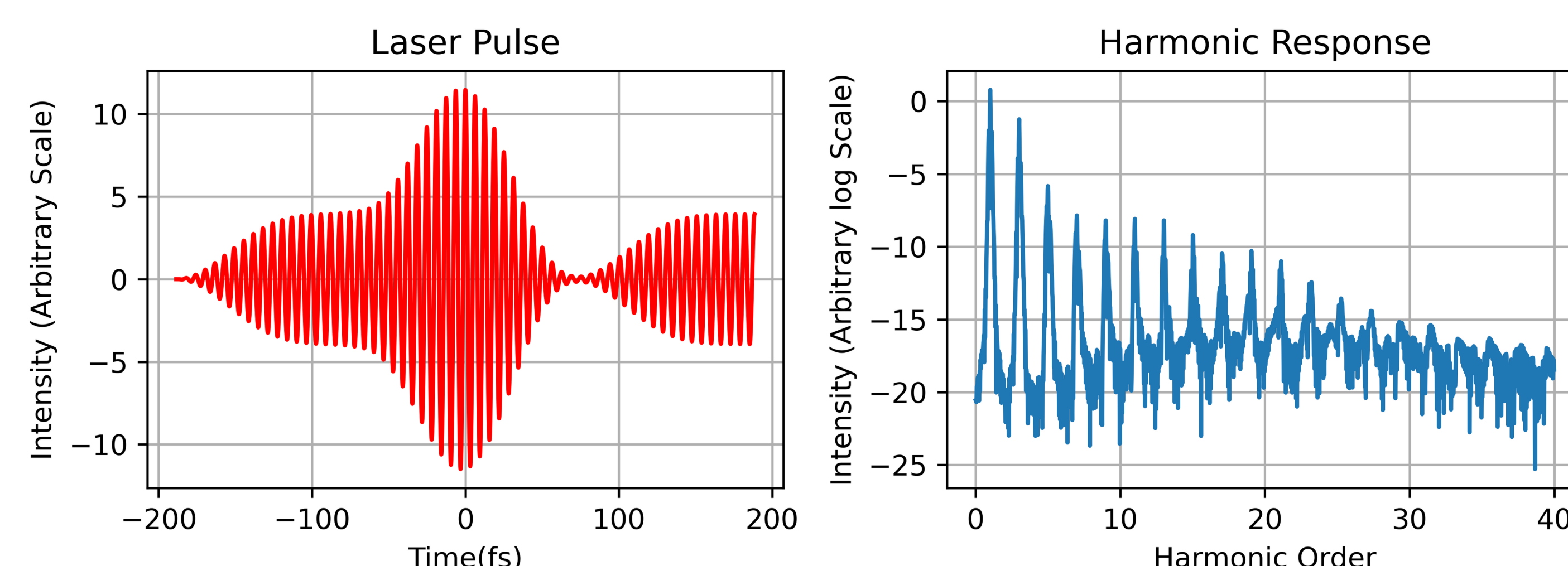
Wavelet Analysis

Another useful tool is wavelet analysis: by applying localised Fourier Transforms to a signal, the time-dependence of the frequency components can be viewed. Using a flatter field with a higher peak intensity, we can observe the short and long trajectories as different arcs in the graph. We can also see that there are two wavelet cycles per laser cycle as expected.



Funky Fields

Due to taking electric field values as an input rather than a function, SNAIL can calculate the response and analyse it for any field you can imagine!



Future Work

The next steps in development will be to implement far field propagation and easy 3D calculations. SNAIL is still a work in progress, so if there is any functionality you think would be helpful, feel free to send me an email at: alex.benitez@kcl.ac.uk

[1] P. B. Corkum Phys. Rev. Lett. **71**, 1994—1997, 1993

[2] Högnér, Maximilian. Master thesis, Ludwig-Maximilians-Universität München, Fakultät für Physik, 2013